

Event Performance Analysis: Honors Program Strategic Audit

Independent Data Analytics Engagement — Higher Education Sector

Tools & methods: R (tidyverse, lubridate, ggplot2), statistical hypothesis testing, data taxonomy design, executive presentation design | **Stakeholders:** Program leadership, full-time staff team

Situation

A university program office ran 50+ student-facing events annually with no structured way to evaluate performance. Records existed only as a single unstructured export — inconsistent date formats, uncategorized free-text titles, location data fragmented across 30+ spelling variants, and missing capacity fields for roughly a quarter of entries. Leadership needed a defensible, data-driven view of program performance ahead of annual planning, without the resources to commission a formal analytics build.

Approach

I scoped, structured, and executed the engagement end-to-end. The open-ended ask was broken into discrete, testable sub-questions before any analysis began. I designed a reusable data taxonomy from unstructured input — a 12-category event classification system and venue-consolidation logic — to make four years of inconsistent data analyzable. I applied statistical testing rather than visual inference alone to confirm or rule out hypotheses, explicitly separating directional findings from statistically robust ones to build credibility into the recommendations. Current-period results were validated against a historical baseline to guard against over-fitting conclusions to a single semester. The engagement closed with two deliverables built from one analytical base: a methodologically transparent report for leadership, and an executive-style presentation for a full staff offsite.

Scope

4 years / 97 records cleaned · 12-category taxonomy built from unstructured text · 6 analytical dimensions evaluated · 2 audience-specific deliverables produced

Skills Demonstrated

Structuring ambiguous problems into testable frameworks · end-to-end data pipeline design · statistical reasoning on limited samples · translating quantitative findings into decision-ready recommendations · tailoring analysis for technical vs. executive audiences · independent ownership from scoping through delivery.

Specific findings and operational details have been omitted to respect the privacy of the partner organization. This summary reflects process and methodology, not internal performance data.